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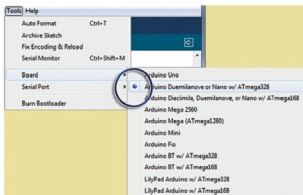


Fig. 6: Screenshot of Arduino IDE

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno or Nano V3
T1	- BC547 npn transistor
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 120-ohm
R2	- 100-ohm
VR1, VR2	- 10-kilo-ohm preset
Miscellaneous:	
LCD1	- 16×2 LCD display
CON1	- 2-pin connector
CON2	- 3-pin connector
CON3	- 6-pin connector
CON4	- 16-pin connector
	- Soil moisture sensor
	- 5V DC regulated USB port power supply

into the soil affect calibration.) Link the sensor head to the rest of the hardware to complete the project.

Rest of the hardware is a combination of Arduino microcontroller and a solid-state display panel. For compactness, Arduino Nano V3 platform is used to drive the 16x2 LCD panel. Though a 6V battery pack is used to energise the hardware setup, you can also use external DC power in 7V - 12V range at V_{DD} of Board1 without any problem. The 10-kilo-ohm preset (VR2) is included to adjust the contrast level of the display. Similarly, the 100-ohm resistor (R2) limits the operating current of the backlight lamp inside the display panel.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

After building the hardware, load the code (sketch) to the microcontroller, that is, copy the sketch, paste it into Arduino IDE, compile and upload. It should work at first try and you should get the message shown in Fig. 3 on your LCD screen.

An actual-size, single-side PCB for the soil moisture meter is shown in Fig. 4 and its component layout in Fig. 5. Enclose the PCB in a suitable box.

Connectors CON1, CON2, CON3 and CON4 are for connecting the sensor probe, analogue input, board 1 and LCD1, respectively.

These days the cheapest USB-to-serial chip on the market is CH340G, and most of the reasonably-priced Arduino-compatible Nano V3 microcontroller modules (with USB mini-B connection) are wired with CH340G USB-to-serial chip on board.

Arduino Nano V3 can be powered via the USB mini-B connection or 5V regulated external DC power supply at 5V pin on Board1.

CH340G chip on Arduino Nano V3 is only powered if the board is being powered over USB. As a result, when running on external (non-USB) power, 3.3V output (supplied by CH340 chip) is not available. Arduino Nano V3 can be programmed with Arduino software (IDE). Just select Arduino Duemilanove or Nano w/ ATmega328 from Tools→Board Menu, as shown in Fig. 6.

Driver for Arduino Nano V3 clone is available at www.electronic-componentsonline.co.in/downloads/ch341ser.zip

Lab note. The project was tested at EFY using Arduino Uno board. **EFY**



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